

ClimateWins

Evaluating Machine Learning Algorithms for Weather Prediction in Mainland Europe



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Objective and Thought Experiments

Objective:



Thought experiments:

Finding new patterns in weather changes over the last 60 years.

Identifying Weather Patterns Outside the Regional Norm in Europe

Generating Possibilities for Future Weather Conditions Over the Next 25 to 50 Years

To use machine learning algorithms for categorizing and predicting weather in mainland Europe.

Machine Learning Options



Random Forest

An ensemble learning method that constructs multiple decision trees and merges their results for improved accuracy and stability.



Hierarchical Clustering

A method of cluster analysis that builds a hierarchy of clusters by either merging or splitting them successively.



Recurrent Neural Network (RNN)

A neural network designed for sequential data, where connections between nodes form a directed graph along a temporal sequence, allowing it to capture temporal dependencies.



Convolutional Neural Network (CNN)

A type of deep learning model particularly effective for analyzing visual data, using convolutional layers to detect patterns and features.



Generative Adversarial Network (GAN)

A deep learning model where two neural networks, a generator and a discriminator, compete to produce increasingly realistic synthetic data.

Thought Experiment 1 - Finding New Patterns in Weather Changes Over the Last 60 Years

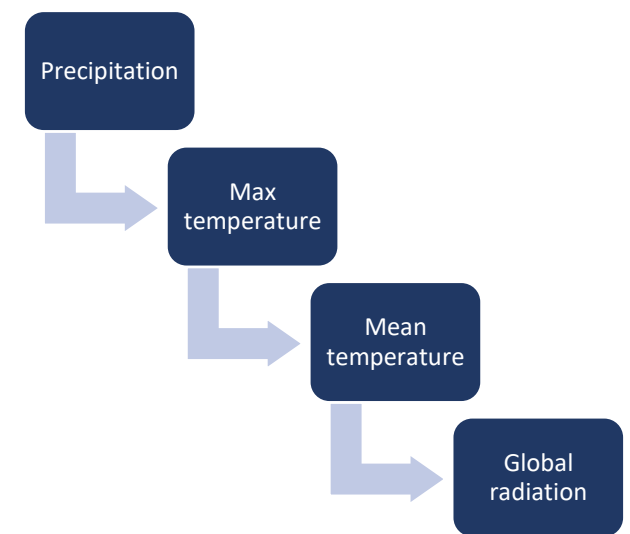
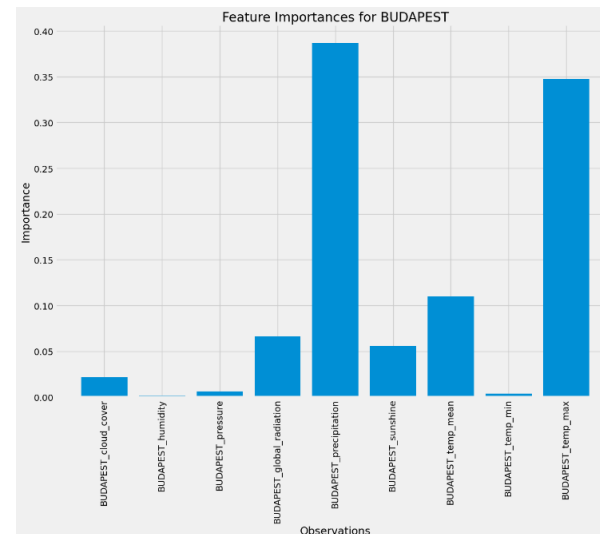
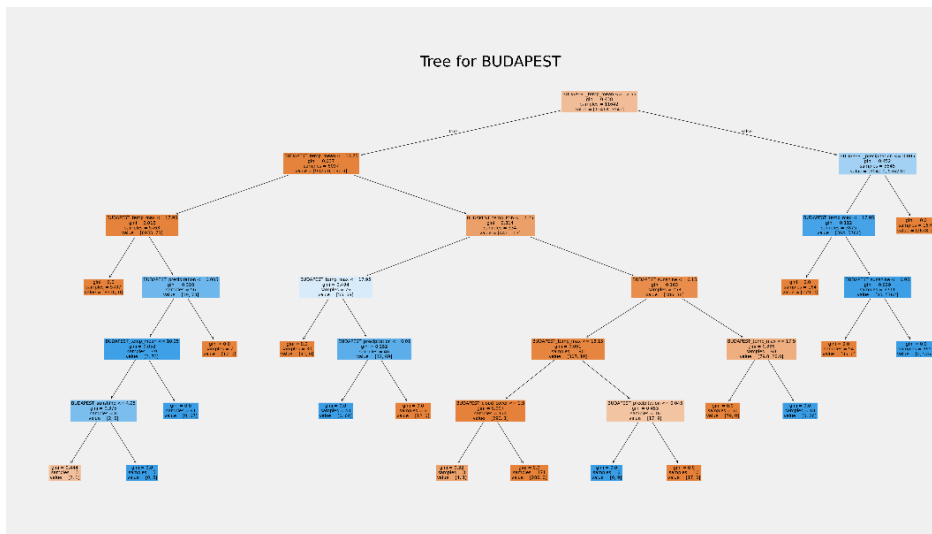


•**Random Forest:** To analyze feature importance and identify key factors influencing weather changes.



•**Data Needed:** Historical weather data, temperature, wind speed, precipitation, and other relevant variables.

Random forest add an impressive accuracy rate of 100% when each weather station was isolated, making it a strong foundation for understanding the key factors!



Thought Experiment 2 - Identifying Weather Patterns Outside the Regional Norm in Europe



•**Hierarchical Clustering:** To group weather patterns, identify outliers and assert regional norms.

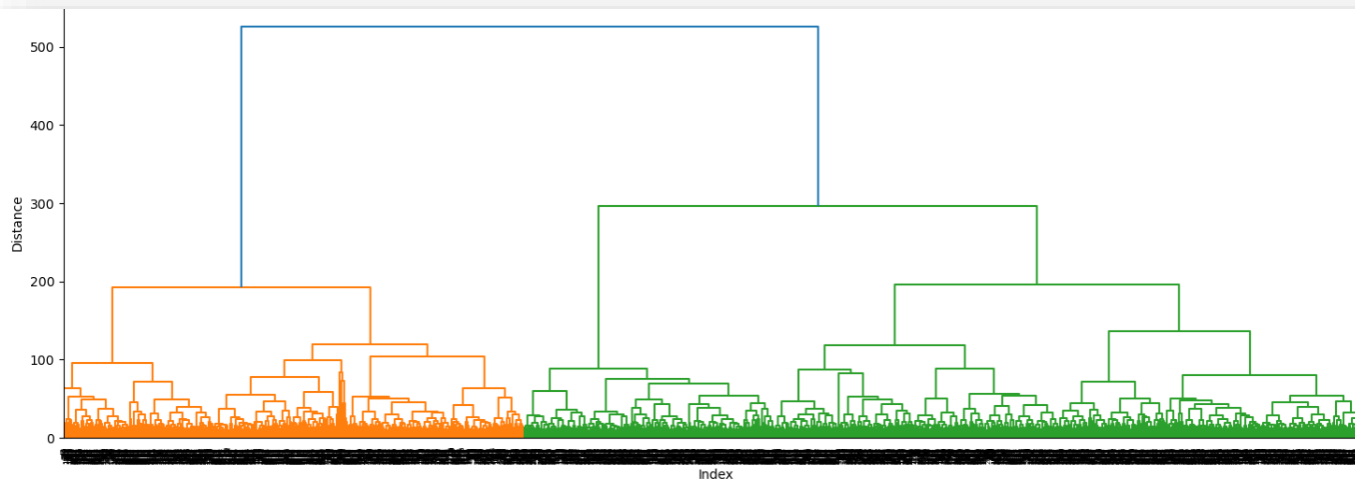


•**CNN:** To analyze spatial patterns in weather data, such as satellite imagery or webcam live images.

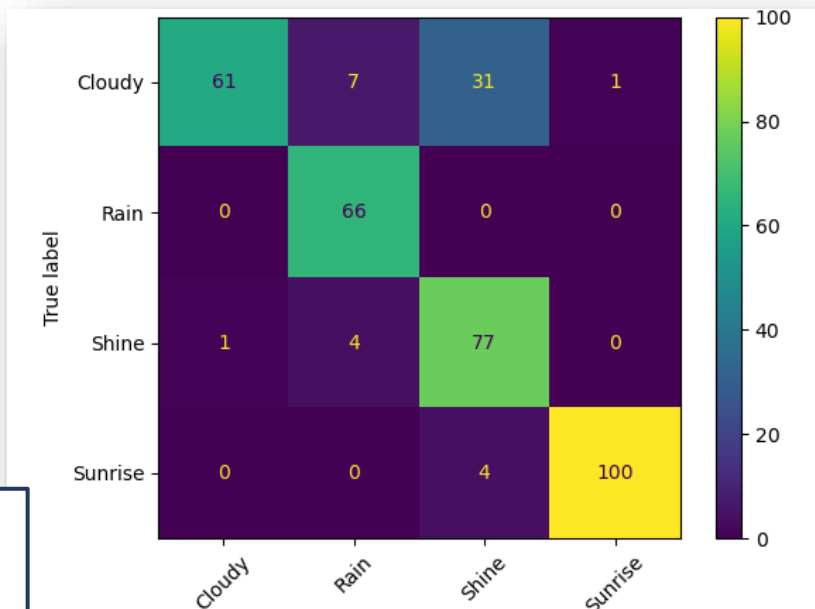


•**Data Needed:** Historical weather data, satellite imagery, regional climate data.

Dendrogram from a hierarchical clustering on weather observations



Confusion matrix of the weather recognition



The CNN did a great job at recognizing the weather conditions from pictures, with some place for improvement left on cloudy vs shiny days since they both contain pictures of the sky.

Thought Experiment 3 - Generating Possibilities for Future Weather Conditions Over the Next Years



- **GANs:** To generate synthetic weather data and simulate future conditions.



- **RNN:** To predict long-term weather trends and patterns.

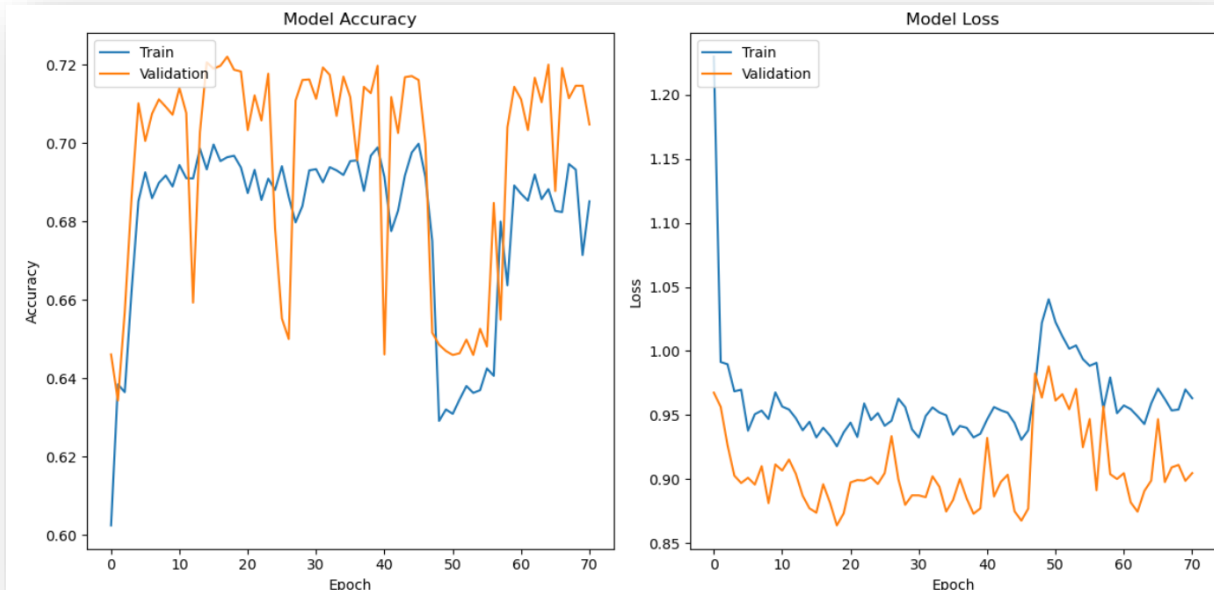


- **Data Needed:** Historical weather data, climate models, socio-economic data.

The RNN had a great optimization curve, from about 4% to 70% accuracy with a simple bayesian optimization! It still has areas where the model struggles to differentiate between similar weather patterns.

216/216 — 2s 8ms/step

Pred	LJUBLJANA	DUSSELDORF	MUNCHENB	SONNBLICK	VALENTIA
True					
LJUBLJANA	4371	71	1	0	4
DUSSELDORF	821	473	5	2	4
MUNCHENB	144	97	2	1	1
MADRID	50	43	2	1	0
STOCKHOLM	20	12	0	0	0
SONNBLICK	57	38	2	2	0
KASSEL	7	6	0	0	0
BUDAPEST	56	12	1	2	0
DEBILT	7	2	0	0	0
VALENTIA	428	106	6	2	4
BASEL	7	1	0	0	0
BELGRADE	2	5	0	0	0
MAASTRICHT	0	4	0	0	0
OSLO	3	0	0	0	0





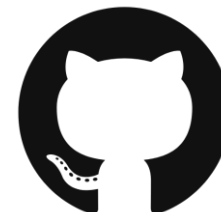
Recommendations

- ⚡ **Thought Experiment 1** (Finding New Patterns in Weather Changes) shows the highest potential for immediate implementation and forms a strong foundation for other analyses.
- ⚡ **Suggested Algorithm Implementation Plan:**
 - Implement and optimize Random Forest models for each weather station
 - Develop initial hierarchical clustering system
 - Deploy CNN for pattern recognition from satellite imagery
 - Integrate RNN for temporal pattern analysis
 - Develop initial GAN models for weather simulation
 - Create composite models combining multiple algorithms
 - Implement real-time prediction systems
- ⚡ **Data requirements:**
 - Extended Historical Data (altitudes, wind direction, atmospheric pressure)
 - Satellite imagery archives
 - Ocean temperature
 - Urban development and land use changes

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**Thank you for your
kind attention !**

For any question, please don't
hesitate to contact me:



All scripts are available on my
dedicated GitHub repository:

